Thermal Conductivity Measurements on Suspended Diamond Membranes Using Picosecond and Femtosecond Time-Domain Thermoreflectance

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ABSTRACT

We report on the room temperature in-plane thermal conductivity measurements on a 1-micrometer thick suspended diamond membrane grown by chemical vapor deposition using two different time domain thermoreflectance (TDTR) setups. The first setup is at Stanford University and the second is at Georgia Institute of Technology. Despite the differences between the two setups and the difficulty associated with diamond membranes thermal measurements, we demonstrate excellent repeatability from each setup and a very good agreement between the two setups. The paper outlines steps considered by both groups to minimize the measurement uncertainty and achieve such agreement. The measurement results show that the thermal conductivity displays a large variability across the membrane. The sensitivity and uncertainty analyses suggest that this variability could be a result of the nonuniformity in the diamond and aluminum coating thicknesses across the sample.

KEY WORDS: thermal conductivity, CVD diamond, time domain thermoreflectance.

INTRODUCTION

The integration of diamond thin films as thermal spreaders in electronic and optoelectronic applications promises for improved thermal management solutions given their high thermal conductivity [1]. However, the measurement of thermal conductivity of diamond films is challenging since the sensitivity of such measurements decreases with increasing thermal conductivity regardless of the considered technique [2]. The thermal anisotropy of thin diamond films makes the measurement even harder by introducing an extra unknown into the measurement and analysis. In literature, there exists a wide variation in thermal conductivity of thin diamond films. However, it is not clear whether this discrepancy is a result of differences in the structure and composition of measured samples or due to the measurement uncertainty in the utilized technique. Time-domain thermoreflectance (TDTR) has been one of the main methods used to study thermal transport across diamond films [3]–[6]. The technique is based on creating a periodic heating event using a modulated laser beam on the surface of a metal coated sample and measuring the thermal response of the sample through another beam. The spatial resolution of TDTR is limited by the laser spot size which is usually of the order of few micrometers. In order to produce higher confidence in TDTR measurement results, a careful understanding of the measurement sensitivity, sources of error, and various contributors to the uncertainty are required. In this paper, we utilize two different TDTR systems, one at Stanford University (Stanford) and the other at Georgia Institute of Technology (GT) and measure the thermal conductivity of a 1 μm thick suspended diamond membrane grown by chemical vapor deposition. Given how challenging these measurements are and in order to improve the agreement between Stanford and GT, multiple measurements were performed by both groups and careful error/uncertainty analysis was considered. Both groups shared as much of details on the measurement procedure and fitting parameters in order to minimize discrepancies in the data analysis. The Diamond film is originally grown on a silicon substrate before a small area of the substrate is etched off. A schematic of the sample configuration is given in Fig. 1. The rms roughness of the diamond surface is 19 nm and was measured by Atomic Force Microscope (AFM). The diamond membrane thickness of 1 μm is the nominal thickness from deposition.

EXPERIMENTAL METHODS

We use time domain thermoreflectance to measure the in-plane thermal conductivity of our diamond sample. A schematic of the main components of each of Stanford and GT systems is given in Fig. 1. The TDTR system at Stanford (GT), is centered around a Nd:YVO4 (Ti-Sapphire) laser of f_s = 82 MHz (80 MHz) pulse train with pulse widths of $\approx$10 ps (100 fs) and central wavelength of 1064 nm (800 nm). The laser output is split into separate pump and probe beams. The pump beam is modulated at a frequency f_{mod} by an electro-optic modulator. The probe is delayed by a mechanical delay stage that gives a time delay of up to 3.6 ns (7.5 ns). The pump beam is converted from 1064 nm (800 nm) to 532 nm (400 nm) using a LiF (BiBO) second harmonic generation crystal. The pump and probe beams are then coaxially focused on the sample down to $1/e^2$ pump and probe radii of r_0 = 4.86 μm (5 μm) and r_1 = 2.97 μm (2.1 μm), respectively. The diamond membrane is coated with 87 nm of aluminum which acts as a temperature transducer.

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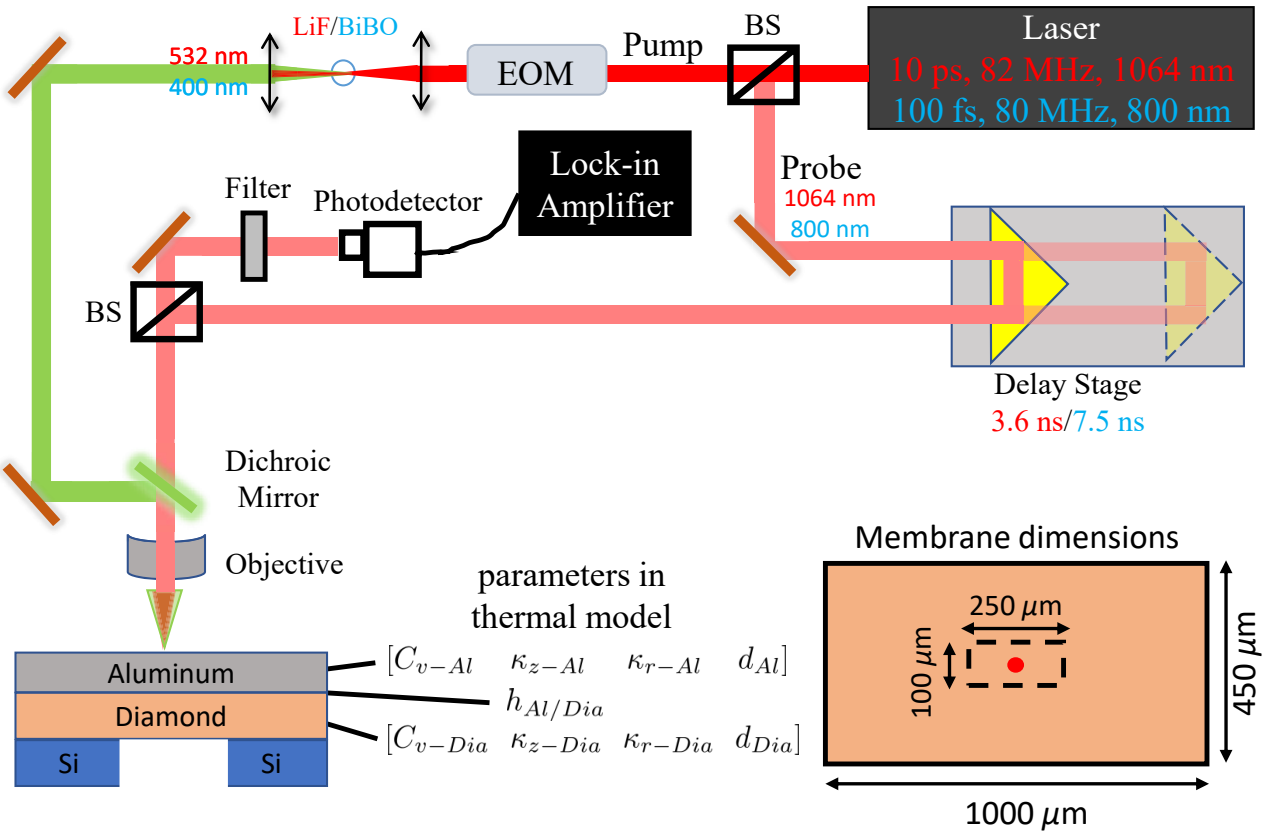


Fig. 1: A schematic of the main components of TDTR. The text labels in red (blue) correspond to the components of the picosecond (femtosecond) TDTR setup at Stanford (GT). The lower right corner shows the dimensions of the diamond membrane. The dashed rectangle corresponds to the area over which in-plane thermal conductivity maps were obtained (see Fig. 5).

The probe beam is retroreflected into a photodetector to monitor the Al thermoreflectance response caused by the surface temperature oscillations. An optical high-pass filter is mounted in front of the photodetector to prevent any reflected pump light from dominating the thermoreflectance signal. We use a lock-in amplifier to monitor the ratio of the in-phase to out-of-phase signal and collect the data as a function of the time delay between the pump and probe. The thermoreflectance data is then related to a thermal model that contains a solution to the heat diffusion equation in a multilayer stack and accounts for radial heat spreading and accumulation effects from the laser pulses [7], [8].

The dimensions of the diamond membrane are given in the lower right corner of Fig. 1. The dashed triangle in the middle is the target of a thermal conductivity mapping in order to investigate the uniformity and repeatability of the this property across the membrane.

SENSITIVITY ANALYSIS

The parameters in our thermal model include the aluminum (diamond) cross-plane and in-plane thermal conductivities, $\kappa_{z-Al(Dia)}$ and $\kappa_{r-Al(Dia)}$, respectively, volumetric heat capacity of aluminum (diamond), $C_{v-Al(Dia)}$, aluminum (diamond) thickness, $d_{Al(Dia)}$, the Al/diamond thermal boundary resistance, TBR (or alternatively, Al/diamond thermal boundary conductance, $h_{Al/Dia}$), r_0, r_1, f_{mod}, and f_s. The sensitivity of the ratio signal $(-V_{in}/V_{out})$ to a certain parameter x in the TDTR model is calculated using [9]:

$$S_x = \frac{\partial \log\left(-V_{in}/V_{out}\right)}{\partial \log x}. \tag{1}$$

Figure 2 shows S_x for the main parameters that affect our TDTR analysis calculated using the values listed in Table I. In our TDTR measurement, we are usually able to extract two parameters at the same time, κ_{r-Dia} and TBR in this case, while the others are inputted into the thermal model. Clearly, the measurement is highly sensitive to d_{Al}, d_{Dia}, r_0, C_{v-Dia}, and C_{v-Al}. Spot sizes, r_0 and r_1, were measured using knife edge technique at Stanford and a beam profiler at GT right before the TDTR measurements to ensure an up-to-date value for the spot sizes. We assume literature value for C_{v-Al} [10] and measure d_{Al} by picosecond acoustics [11] using the GT system. We measure the electrical resistivity of the Al film by four-point probe to back up κ_{r-Al} from Wiedmann-Franz law. We note that this is a common practice in TDTR measurements and the obtained Al thermal conductivity has shown

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a very good agreement with independent thermoreflectance measurements on Al films down to thicknesses as low as 30 nm [12]. The Al film is also assumed to be isotropic so $\kappa_{z-Al} = \kappa_{r-Al}$. κ_{z-Dia} is obtained by performing TDTR measurement at a higher f_{mod} using a large spot size on a location where the silicon substrate is not etched off (see inset of Fig. 3). A value of 180 $\mathrm{Wm^{-1}K^{-1}}$ was obtained. Because of the high Debye temperature of diamond, its heat capacity is sensitive to the measurement temperature. As a result, the steady-state temperature rise due to the absorbed laser powers during TDTR measurements needs to be taken into consideration. Table I shows the values for the pump and probe powers, P_0 and P_1, respectively, incident on the sample. The absorbed powers are calculated using literature values of the absorptivity of Al to the wavelengths of the pump and probe. The steady-state temperature rise is then estimated from the TDTR model in the limit where the time delay between the pump and probe goes to infinity using the parameters in Table I and was to found to be 4 - 8 K. C_{v-Dia}, along with TBR and κ_r, are obtained by doing simultaneous TDTR fitting of a set of measurements taken at three different frequencies using the GT system. The temperature dependence of C_{v-Dia} is then obtained by repeating the same set of measurements using different laser powers. We found that for our specific diamond sample, $C_{v-Dia}(T) = 1.095 \times 10^4 T - 1.504,346 \times 10^6$ in J $\mathrm{m^{-3}K^{-1}}$ where T is the temperature in Kelvins. We note that fitting the literature values [13] for bulk diamond heat capacity around room temperature gives $C_{v-Dia-bulk}(T) = 1.558 \times 10^4 T - 2.785,2514 \times 10^6$ which for T = 302 K (our measurement temperature including the steady state temperature rise) returns a heat capacity of 1.92 $\times$ 10^6 J $\mathrm{m^{-3}K^{-1}}$ whereas the obtained heat capacity of our thin diamond membrane is 1.8 $\times$ 10^6 J $\mathrm{m^{-3}K^{-1}}$. Once, $C_{v-Dia}(T)$ is determined, the subsequent measurements are done at a single frequency with κ_{r-Dia} and TBR being the only unknowns. These two parameters are varied to fit the data to the thermal model using a nonlinear least square minimization routine.

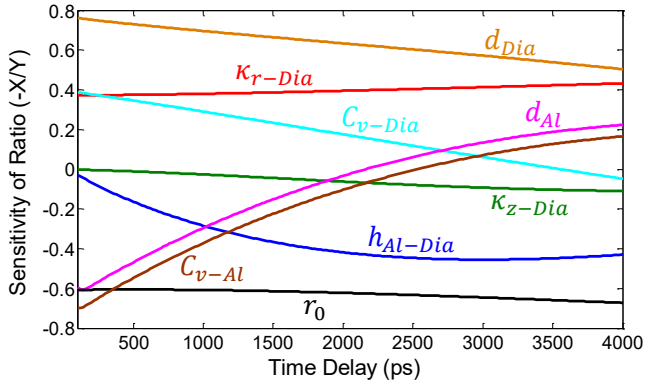


Fig. 2: Sensitivity of the ratio to different parameters in the thermal model calculated using Eq. 1 and the values in Table I.

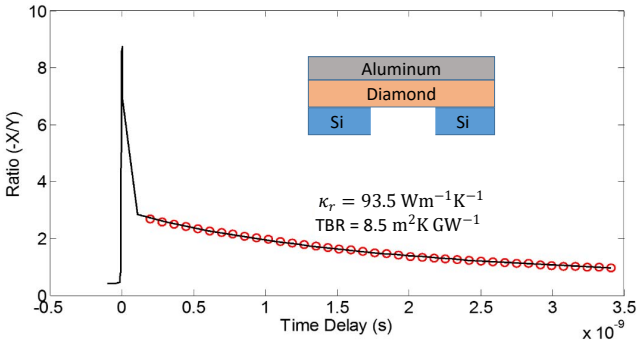


Fig. 3: Representative TDTR scan performed at the center of the membrane at 2.2 MHz. The labels are the best fit values for in-plane thermal conductivity and thermal boundary resistance between aluminum and diamond. Nine TDTR measurements were performed at the same location yielding $\kappa_r = 93.6 \pm 1\,\mathrm{Wm^{-1}K^{-1}}$ and TBR = 8.62 $\pm$ 0.24 $\mathrm{m^2\ K\ GW^{-1}}$.

THERMAL CONDUCTIVITY RESULTS

We performed nine TDTR measurements on the center of the membrane to obtain κ_{r-Dia} and TBR. Figure 3 shows a representative scan at f_{mod} = 2.2 MHz taken at Stanford. The nine TDTR measurements yielded the following average values: κ_r= 93.6 $\pm$ 1 $\mathrm{Wm^{-1}K^{-1}}$ and TBR = 8.62 $\pm$ 0.24 $\mathrm{m^2\ K\ GW^{-1}}$ where the uncertainty is the standard deviation of the nine measurements. Since these measurements were taken at the same location, the uncertainty is tied directly to the signal-to-noise ratio and the sensitivity of the measurement. To investigate the uniformity of thermal conductivity across the membrane, we perform a TDTR mapping of the membrane. For this purpose, the time delay between the pump and the probe is fixed at 100 ps and the sample is mounted on an automated (manual) mechanical XY stage at Stanford (GT). An area of 250 μm $\times$ 100 μm in the center of the membrane

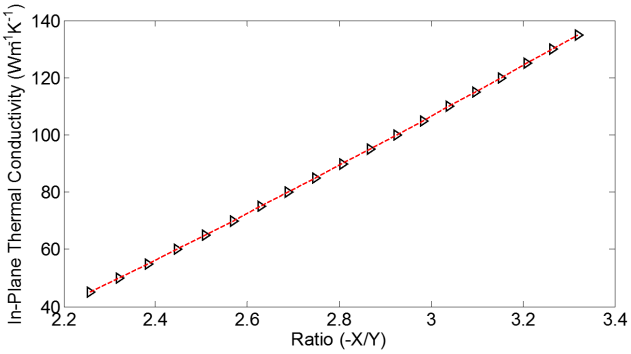


Fig. 4: Converting TDTR ratio signal at 100 ps to in-plane thermal conductivity. The data points are generated by evaluating the TDTR ratio model at the parameters listed in Table I and the average TBR value from the nine full TDTR scans. The dashed line represents a third order polynomial fit.

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TABLE I: List of parameters used in the TDTR model. The x and y values in the $x \rightarrow y$ expressions in parenthesizes represent the typical uncertainty value for this specific parameter and the corresponding propagated error into κ_{r-Dia}, respectively.

Physical Properties				
Material	C_v (MJm^{-2}K^{-1})	κ_z (Wm^{-1}K^{-1})	κ_r (Wm^{-1}K^{-1})	d (nm)
Al	2.43 (2.5%→5%)	140	140 (10%→1.9%)	87 (4%→7%)
Diamond	1.80 (10% →10%)	180	92.7	1000 (5%→9.5%)
Laser Parameters				
f_{mod} (MHz)	r_0 (μm)	r_1(μm)	P_0 (mW)	P_1(mW)
2.2	4.86 (2.5%→4.3%)	2.97 (2.5%→1.6%)	10.1	1

is scanned in steps of 10 μm while monitoring the ratio signal on the lock-in amplifier. Based on our TDTR model and using the parameters listed in Table I, we compute the ratio for different input values of the in-plane thermal conductivity. The computed ratio values are then fitted to a third-order polynomial which is used to convert the measured ratio signal at 100 ps to in-plane thermal conductivity. In this way, we eliminate the need to perform a full TDTR scan at each location on the membrane to obtain a map. Figure 4 shows the conversion of the ratio signal to in-plane thermal conductivity. Figures 5a- 5c show the results of three automated thermal conductivity maps obtained at Stanford on the same area of the membrane. Each map consists of a total of 286 pixels. An excellent level of repeatability can be visually noted when comparing the maps. In addition, the three maps produce nearly the same mean, standard deviation, and range values for the entire area. We note that the thermal conductivity maps were obtained assuming a single value for TBR, the average value from the nine full TDTR scans. While it is expected that TBR will vary across the membrane due to the variation in roughness and interfacial morphology, the sensitivity of the ratio signal to TBR at 100 ps is negligible as can be observed in Fig 2. A similar argument can be made with respect to κ_{z-Dia}. A similar map is obtained by GT (not shown here) using a manual micropositioning stage where a total of 75 pixels were covered in the same membrane area and in steps of 10 μm. The average value for κ_{r-Dia} from all maps obtained at Stanford is 92.7 $\pm$ 7.5 Wm^{-1}K^{-1} and that obtained by GT is 87.7 $\pm$ 10.4 Wm^{-1}K^{-1} where uncertainties are the standard deviations of all measurements. This excellent agreement between the two systems and between the maps and the full TDTR scans demonstrates the possibility of achieving agreement on such challenging type of measurement.

VARIABILITY IN THERMAL CONDUCTIVITY

Figures 5 shows a large variability in the measured in-plane thermal conductivity with values varying between ≈80 and 110 Wm^{-1}K^{-1} across the membrane area. To identify the source of this variability we first consider the uncertainty of the measurement to the various input parameters to the thermal model. The values in parenthesizes for some of the parameters in Table I represent typical uncertainties of these parameters and the corresponding error they cause in κ_{r-Dia} for a specific pixel (notation $x\% \rightarrow y\%$). For example, a 4% error in d_{Al}, and 10% error in d_{Dia} will independently lead to 7% and 10% errors in the measured κ_{r-Dia}, respectively. Moreover, a 10% error in C_{v-Dia} and 2.5% in r_0 will result in 10% and 4.3% errors in κ_{r-Dia}, respectively. These large uncertainties suggest that the nonuniformity in κ_r in Figure 5 could be a result of nonuniformity in some of the input parameters. Given that the rms roughness of the diamond film is 19 nm, a nonuniformity in either or both of d_{Dia} and d_{Al} could be the cause for this large variation in thermal conductivity. Given the large measurement area (250 μm $\times$ 100 μm) obtaining a more accurate measurement of d_{Al} and d_{Dia} using AFM or TEM is not practical. Alternatively, the use of an ellipsometer would help identify the level of this nonuniformity across large areas of the membrane.

Prior and after the measurements, several calibrations were performed to ensure that the results are not due to errors in the experimental procedure. At Stanford, we obtain a set of three thermal conductivity maps on aluminum coated sapphire substrate of the same area as the maps shown in Fig. 5. At GT, line scans are obtained on an aluminum coated silicon substrate. In both cases, the standard deviation was of the order of 2-3%. The rms surface roughness of these two calibration samples was of the order of 3 nm whereas that of the diamond sample was 19 nm. In extreme cases, the sample roughness can affect the shape of the reflected probe spot and the profile of the absorbed pump causing a deviation from the assumed Gaussian intensity profile and leading to a large variation in the measured property. While performing the measurements on the diamond membrane and the calibration samples, we recorded the DC voltage of the photodetector due to the reflected probe signal. The variation of this voltage across the maps can be used to evaluate the variation in the reflectivity and its influence on the measured thermal conductivity. No correlation between κ_{r-Dia} and the reflectivity was found eliminating this factor as a possible cause for the observed nonuniformity.

We also note that the total travel distance of the automated XY stage during the measurement (250 μm), is too short to cause defocusing of the sample. This was checked by

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performing full TDTR scans at locations that are 250-500 μm apart on the calibration samples and obtaining the same thermal conductivity results.

TEM images on our diamond membrane sample (not presented here) demonstrate that the average grain structure is of the order of 100 nm. This is much smaller than our laser spot sizes (6 and 10 μm). Therefore, the variability in κ_{r-Dia} can not be a result of grain size variation across the membrane.

The total uncertainty in κ_{r-Dia} at a specific pixel can be estimated from:

$$\Delta\kappa_{r-Dia} = \sqrt{\delta\kappa_{exp}^2 + \sum_i (\delta\kappa_x)^2} \qquad (2)$$

Where $\delta\kappa_{exp}$ is the uncertainty in κ_{r-Dia} due to experimental noise and is $\approx 2\%$ and $\delta\kappa_x$ is the uncertainty in κ_{r-Dia} due to the uncertainty in the input parameter x as given in Table 1. The final uncertainty is found to be 16%. We note that this uncertainty can be significantly reduced if d_{Al} and d_{Dia} were obtained more accurately.

SUMMARY

We presented in-plane thermal conductivity measurements on 1 μm CVD diamond membrane. Our measurements demonstrated a very good agreement between two TDTR systems, one at Stanford University and the other at Georgia Institute of Technology. The results showed a large variation in the thermal conductivity across a membrane area of 250μm$\times$100μm. The uncertainty and sensitivity analyses suggest that this large variation is due to a nonuniformity in the Al and diamond thicknesses.

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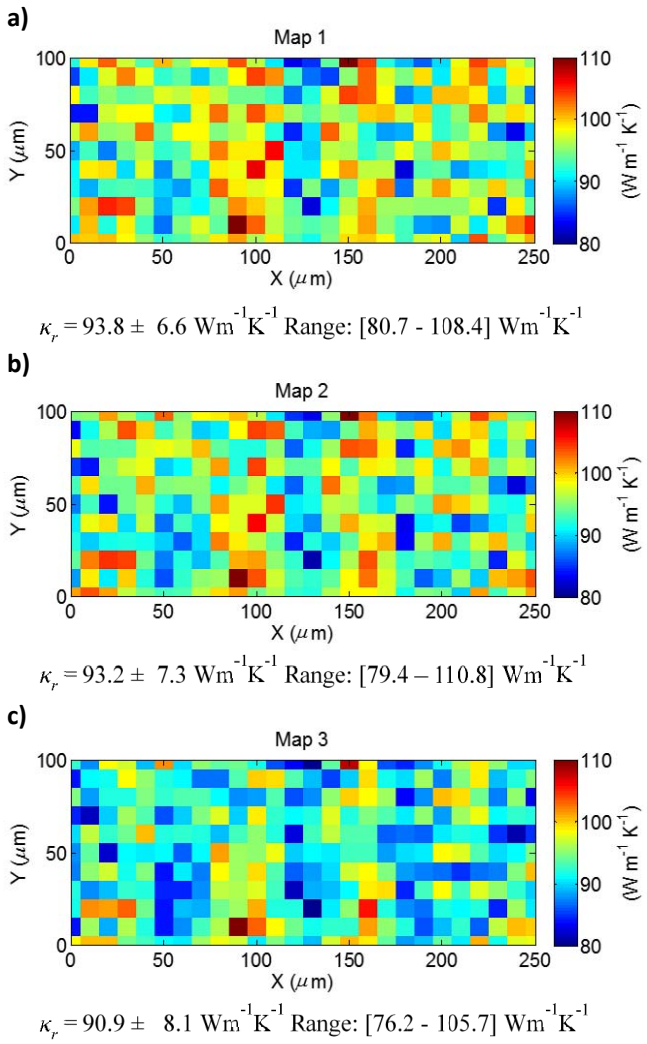


Fig. 5: (a-c) Stanford results for automated in-plane thermal conductivity maps at a pump-probe time delay of 100 ps and using the parameters listed in Table I. The mean value of the entire map and range of thermal conductivities are given below each map.

References

[1] K. E. Goodson, "Thermal conduction in nonhomogeneous cvd diamond layers in electronic microstructures," *Journal of Heat Transfer*, vol. 118, no. 2, pp. 279–286, 1996.

[2] S. Rossi, M. Alomari, Y. Zhang, S. Bychikhin, D. Pogany, J. Weaver, and E. Kohn, "Thermal analysis of submicron nanocrystalline diamond films," *Diamond and Related Materials*, vol. 40, pp. 69 – 74, 2013.

[3] A. Sood, J. Cho, K. D. Hobart, T. I. Feygelson, B. B. Pate, M. Asheghi, D. G. Cahill, and K. E. Goodson, "Anisotropic and inhomogeneous thermal conduction in suspended thin-film polycrystalline diamond," *Journal of Applied Physics*, vol. 119, no. 17, p. 175103, 2016.

[4] C. Monachon and L. Weber, "Effect of diamond surface orientation on the thermal boundary conductance between diamond and aluminum," *Diamond and Related Materials*, vol. 39, pp. 8 – 13, 2013.

[5] G. T. Hohensee, R. Wilson, and D. G. Cahill, "Thermal conductance of metal–diamond interfaces at high pressure," *Nature communications*, vol. 6, 2015.

[6] E. Bozorg-Grayeli, A. Sood, M. Asheghi, V. Gambin, R. Sandhu, T. I. Feygelson, B. B. Pate, K. Hobart, and K. E. Goodson, "Thermal conduction inhomogeneity of nanocrystalline diamond films by dual-side thermoreflectance," *Applied Physics Letters*, vol. 102, no. 11, p. 111907, 2013.

[7] D. G. Cahill, "Analysis of heat flow in layered structures for time-domain thermoreflectance," *Review of Scientific Instruments*, vol. 75, p. 5119, Nov. 2004.

[8] A. J. Schmidt, X. Chen, and G. Chen, "Pulse accumulation, radial heat conduction, and anisotropic thermal conductivity in pump-probe transient thermoreflectance.," *Review of Scientific Instruments*, vol. 79, p. 114902, Nov. 2008.

[9] R. M. Costescu, M. A. Wall, and D. G. Cahill, "Thermal conductance of epitaxial interfaces," *Physical Review B*, vol. 67, p. 054302, Feb 2003.

[10] C. Y. Ho, R. W. Powell, and P. E. Liley, "Thermal conductivity of the elements," *Journal of Physical and Chemical Reference Data*, vol. 1, no. 2, pp. 279–421, 1972.

[11] C. Thomsen, J. Strait, Z. Vardeny, H. J. Maris, J. Tauc, and J. J. Hauser, "Coherent phonon generation and detection by picosecond light pulses," *Physical Review Letters*, vol. 53, pp. 989–992, Sep 1984.

[12] A. J. Schmidt, R. Cheaito, and M. Chiesa, "Characterization of thin metal films via frequency-domain thermoreflectance," *Journal of Applied Physics*, vol. 107, no. 2, p. 24908, 2010.

[13] A. C. Victor, "Heat capacity of diamond at high temperatures," *The Journal of Chemical Physics*, vol. 36, no. 7, pp. 1903–1911, 1962.

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